LIFTING MAPS FROM BOUNDED PROJECTIVE COMPLEXES THROUGH A COHOMOLOGICAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7136-7143, immediately following the statement at 7120-7134.

Editorial note. Both statement hypotheses are consumed at the precise-vanishing step.

Allowed standard prerequisites (editorial).

- Precise vanishing against the cone. Lemma lemma-precise-vanishing of this chapter, cited at proof lines 7140-7141. i (beta) = 0 in K(A) under the hypotheses on P and alpha.
- Long exact sequence of a representable functor. Lemma lemma-representable-homological of this chapter, cited at proof lines 7142-7143. Hom(P, -) is homological, so a map to L that vanishes composed into the cone triangle lifts to E.

Source reference. Original lemma 064D: sources/stacks-derived/pinned-original-source.tex, source lines 7089–7089. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\beta : P^\bullet \to L^\bullet$ and $\alpha : E^\bullet \to L^\bullet$ be maps of complexes. Let $n \in \mathbf{Z}$. Assume*

(1) *$P^\bullet$ is a bounded complex of projectives and $P^i = 0$ for $i < n$,*
(2) *$H^i(\alpha)$ is an isomorphism for $i > n$ and surjective for $i = n$.*

Then there exists a map of complexes $\gamma : P^\bullet \to E^\bullet$ such that $\alpha \circ \gamma$ and β are homotopic.

Proof. Consider the cone $C^\bullet = C(\alpha)^\bullet$ with map $i : L^\bullet \to C^\bullet$. Note that $i \circ \beta$ is zero by Lemma 064D. Hence we can lift β to $E^\bullet$ by Lemma 0149. □

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